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PYTHON: FUNCTION

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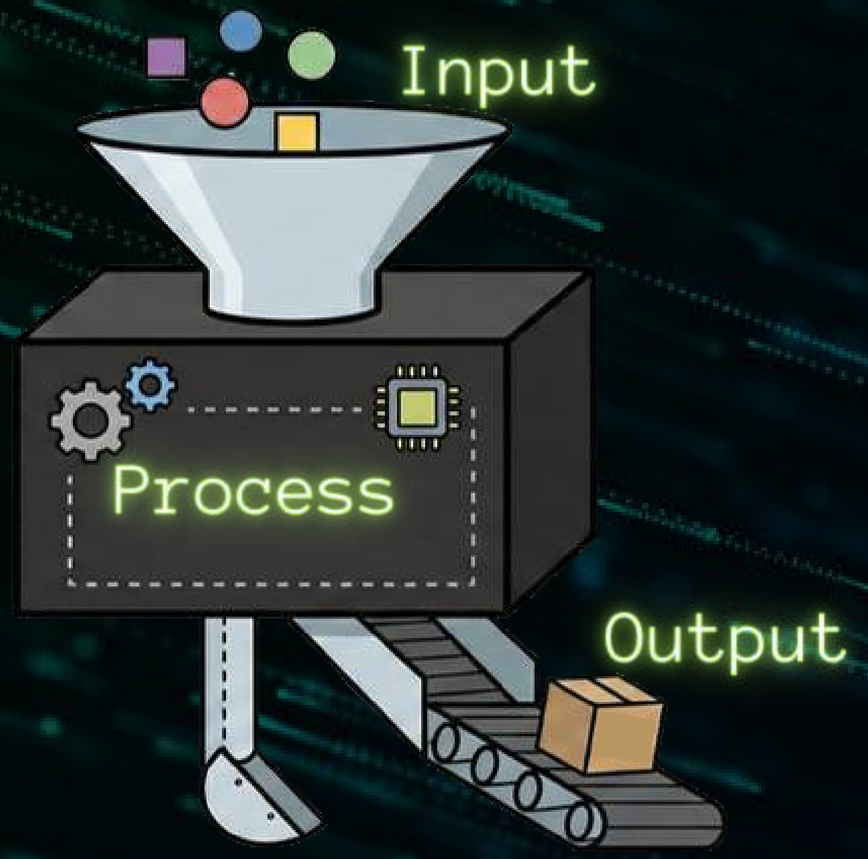
WHAT IS FUNCTION?

A function is a reusable block of code which only runs when it is called.

```
def calculateArea(radius):
    area = 3.1415 * (radius ** 2)
    return area
```

How to use?

```
area = calculateArea(5)
```



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FUNCTION: DRY PRINCIPLE

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DRY: DON'T REPEAT YOURSELF

- If you have to write it twice, it warrants a function
- Centralized logic

Let's demonstrate!!

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FUNCTION: THE ANATOMY

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FUNCTION ANATOMY

Function Definition (def)

- The keyword that tells Python "I am about to build a new tool."
- Names are case sensitive.
- The Rule: It must always be followed by a name, parentheses () and colon :.
- Code: def calculate_total():

FUNCTION ANATOMY

The Body (The Indented Block)

- What is it? The instructions inside the machine.
- The Rule: Python relies on indentation. If you stop indenting, the function ends.

Code:

```
def say_hello():
    print("Hi there!")
```

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FUNCTION ANATOMY

Parameter (The Variable)

- Definition: The variable name used inside the function definition. It is a placeholder.
- Where it lives: Inside the parentheses of the def line.

Argument (The Value)

- Definition: The actual data you send into the function when you use it.
- Where it lives: Inside the parentheses when you call the function.

FUNCTION ANATOMY

The Return Statement

- Definition: The command that ends the function and sends a value back to where it was called.
- Key Concept: Without a return, the function calculates the answer and then throws it in the trash (returns None).

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FUNCTION ANATOMY

SCOPE

Local Variables (The "Private" Space)

- Definition: Variables created inside a function.
- Visibility: Only that specific function can see them.

Global Variables (The "Public" Space)

- Definition: Variables created in the main body of your code (outside any function).
- Visibility: Everyone can see them. Any function can access them.

FUNCTION ANATOMY

Shadowing

What if I have a global variable named x and a local variable named x? Which one does Python use?

Let’s experiment

The Rule: "Local always wins."

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FUNCTION: LET’S REFACTOR

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FUNCTION: YOU REFACTOR

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FUNCTION OF PYTHON

Built-in Functions

Let’s experiment len(container):
What it does: Counts how many items are inside.

Use Case: "How many students are in the class?"
max(items) / min(items):

What it does: Finds the highest or lowest value.
Use Case: "Who got the highest score?"
sum(items):

What it does: Adds everything up.
Use Case: "What is the total cost of the cart?"

ACTIVITY

Activity: The Payroll System (Logic)

1. Function: get_pay(hours, rate)
   - Logic:
     - If hours > 40: First 40 are normal. Extra hours are 1.5x (Overtime).
     - If hours <= 40: Just multiply hours * rate.
   - Action: Return the total pay.
2. Function: get_tax(amount)
   - Action: Return 10% of the amount (amount * 0.10).
3. Main Function: payslip(name, hours, rate)
   - Call get_pay to get Gross Pay.
   - Call get_tax to get Tax.
   - Print: "Net Pay for [Name]: $[Gross - Tax]"

Your Task: Run these 2 test cases

- Intern: 20 hrs @ PHP15/hrs (Expected Net: PHP270)
- Dev: 50 hrs @ PHP50/hr (Overtime) (Expected Net: PHP2475)